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Multi-Source Multi-View Graph Domain Adaptation with Hyperbolic Residual Encoding for Cross-Site MDD Identification from rs-fMRI

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Abstract

Cross-site identification of major depressive disorder (MDD) from resting-state functional magnetic resonance imaging (rs-fMRI) is hindered by inter-site distribution shifts and heterogeneous functional connectivity (FC) views. These views capture complementary neural relationships but exhibit distinct site biases and graph topologies, complicating alignment without sacrificing disease-relevant information or cross-view consistency. Existing studies largely treat

multi-view connectome learning and cross-site adaptation separately. To the best of our knowledge, few studies have jointly modeled multiple FC views under multi-source unsupervised domain adaptation for cross-site rs-fMRI-based MDD classification. We construct Pearson correlation, sparse representation, and Granger causality graphs, each encoded by a view-specific graph attention network. Dual-stream adaptive fusion explicitly integrates pairwise cross-view interactions, followed by lightweight hyperbolic residual encoding for curvature-aware representation refinement. Class-wise Cauchy–Schwarz alignment reduces inter-source and source-target discrepancies, complemented by adversarial learning, information maximization, and confidence-aware pseudo-labeling. Across seven unlabeled target domains, our framework achieves 73.60% mean accuracy and 71.90% AUC, demonstrating effective generalization under heterogeneous acquisition conditions. These results highlight the effectiveness of unified heterogeneous-view modeling, curvature-aware refinement, and multi-source domain adaptation for cross-site MDD identification. The source code is at https://github.com/OPUS-Lightphenexx/MM-HyperGDA

Keywords: Domain Adaptation, MDD

1 Introduction

Major depressive disorder (MDD) is among the most prevalent and debilitating mental health conditions worldwide Marx et al. (2023), yet its diagnosis still relies largely on clinical interviews and symptom-based assessment. Resting-state functional magnetic resonance imaging (rs-fMRI) offers a promising means of identifying objective neural biomarkers through intrinsic functional connectivity. However, reliable rs-fMRI-based MDD diagnosis remains challenging.

The practical application of rs-fMRI-based MDD classification is hindered by three increasingly intractable intertwined obstacles. First, data collected from different acquisition sites exhibit severe inter-site distribution shifts, which drastically degrade the generalizability of trained modelsDansereau et al. (2017). Second, the characterization of brain functional connectivity is inherently multi-faceted; different construction methods Bastos and Schoffelen (2016) capture complementary neural relationships, ranging from undirected synchrony to directed causal influences. Critically, these diverse views exhibit distinct site-specific biases and graph topologies Friston (2011). This heterogeneity makes multi-source domain alignment extremely difficult, as naive adaptation strategies risk either erasing disease-relevant, view-specific information or disrupting the essential cross-view consistency among complementary representations. Third, hierarchically organized brain networks may incur representation distortion when modeled exclusively in conventional Euclidean spaces.

While unsupervised domain adaptation can mitigate source–target distribution shifts, extending it simultaneously to multiple sources and multiple functional-connectivity views is substantially more challenging. First, labeled source sites may differ in acquisition protocols, cohort characteristics, and class-conditional distributions; directly pooling them can obscure source-specific structure and induce negative

transfer. Second, Pearson correlation, sparse representation, and Granger causality produce heterogeneous feature distributions and graph topologies. Forcing these views into a common alignment space may suppress disease-relevant view-specific information, whereas adapting them independently may overlook their complementary cross-view dependencies. The joint setting therefore requires coordinating inter-source alignment, source–target adaptation, and cross-view integration while preserving both class-discriminative structure and view diversity.

To address these coupled challenges, we formulate cross-site MDD identification as a unified multi-source, multi-view graph domain adaptation problem. View-specific graph attention encoders preserve the distinct topology of each connectivity construction, while pairwise cross-view interaction and dual-stream adaptive fusion integrate complementary information without collapsing view-specific representations. Class-wise Cauchy–Schwarz alignment jointly reduces inter-source and source–target discrepancies, while adversarial learning, information maximization, and confidence-aware pseudo-labeling provide complementary global and target-structure adaptation signals. Finally, a lightweight post-fusion hyperbolic residual encoder performs curvature-aware refinement of the fused subject-level representation.

In summary, the main contributions of this work are summarized as follows:

1. We formulate cross-site MDD identification as a multi-source, multi-view unsupervised graph domain adaptation problem, where inter-source heterogeneity, source–target shifts, and view-dependent distribution biases are jointly addressed without collapsing class-discriminative structure or view diversity.
2. We develop a unified architecture that preserves view-specific graph structures, explicitly models pairwise cross-view dependencies, and adaptively integrates direct and interaction-based representations before post-fusion curvature-aware refinement.
3. We integrate class-conditional source alignment, source-target distribution matching, domain-adversarial learning, information maximization, and confidence-aware pseudo-labeling into a unified two-stage optimization framework, and evaluate the framework across seven target sites.

2 Related Work

2.1 Cross-Site Domain-Adaptive Diagnosis

Cross-site rs-fMRI analysis is affected by scanner, protocol, and cohort differences that induce substantial domain shifts Dansereau et al. (2017). Unsupervised domain adaptation uses labeled source data and unlabeled target data to learn transferable representations. UFA-Net Fang et al. (2023) combines attention-guided graph learning with distribution alignment for cross-site MDD identification, whereas AUFA Ma et al. (2026) introduces augmentation-based self-optimization to reduce source-model overfitting. In the broader context of cross-site neuropsychiatric diagnosis, H2MSDA Luo et al. (2025) addresses multi-source adaptation for ASD through hierarchical alignment across labeled sources and an unlabeled target. However, these methods primarily rely on a single functional-connectivity construction and therefore do not

explicitly address view-dependent domain shifts or interactions among heterogeneous connectivity representations. This limitation becomes more pronounced when source sites exhibit different class-conditional distributions.

2.2 Multi-View Functional Connectivity Learning

Different connectivity constructions characterize complementary neural relationships. Pearson correlation captures undirected synchronization, sparse representation models reconstruction-based dependence, and Granger causality describes directed temporal predictability. MFCP Ruan et al. (2026) integrates these three views through multiple graph convolutional branches for MDD identification. Related multi-connectome methods also demonstrate the value of heterogeneous graph views in brain-disorder classification Yang et al. (2024); Zhou et al. (2024). Nevertheless, these approaches generally assume matched training and test distributions and thus do not address how multiple connectivity views should be jointly adapted when labeled source sites and an unlabeled target exhibit distinct, view-dependent shifts.

2.3 Hyperbolic Representation Learning

Hyperbolic spaces provide compact representations for data with latent hierarchical organization and have been adopted in neural and graph representation learning Nickel and Kiela (2017); Ganea et al. (2018); Chami et al. (2019). Recent neuroimaging studies have also explored hyperbolic geometry for cross-site alignment Luo et al. (2025). However, existing methods typically operate on a single connectivity construction and do not examine how heterogeneous graph views should be integrated before geometric encoding. In contrast, our framework preserves view-specific structures, models pairwise cross-view dependencies, performs adaptive dual-stream fusion, and applies lightweight Poincaré-ball-constrained residual refinement while jointly reducing inter-source and source–target discrepancies.

3 Materials and Data Preprocessing

3.1 Dataset and Evaluation Protocol

Experiments were conducted on the REST-meta-MDD multi-site resting-state functional magnetic resonance imaging dataset for major depressive disorder identification Yan et al. (2019). Site20 and Site21 were treated as two labeled source domains. Site6, Site11, Site12, Site16, Site19, Site22, and Site25 were independently treated as unlabeled target domains.

For each target-site experiment, all target samples participated in adaptation without ground-truth labels. Target labels were not used in source pretraining, adaptation losses, pseudo-label construction, or parameter optimization, and were used only after training to compute ACC and AUC. The experiments therefore followed a transductive multi-source unsupervised domain adaptation setting.

Each target site was adapted and evaluated independently. Accuracy (ACC) was calculated from the final predicted labels, and the area under the receiver operating characteristic curve (AUC) was calculated from the predicted probabilities. Site-wise

results were reported for all seven target domains. Macro-averaged ACC and AUC were further calculated by assigning equal weight to each target site.

Table 1 Sample statistics of the source and target sites. HC denotes healthy controls.

Site	HC	MDD	Total
Site20	251	282	533
Site21	70	86	156
Site6	15	15	30
Site11	29	32	61
Site12	6	32	38
Site16	31	31	62
Site19	36	51	87
Site22	20	30	50
Site25	63	89	152

3.2 ROI Time Series and Multi-View Network Construction

We used publicly released preprocessed ROI time series from REST-meta-MDD. Following the reported DPARSF pipeline, the first ten volumes were removed, followed by head-motion correction, spatial normalization and smoothing, nuisance-signal regression, and band-pass filtering Yan and Zang (2010); Power et al. (2012); Hallquist et al. (2013). Mean BOLD signals were extracted from 116 AAL regions. We directly loaded these AAL116 signals, z-standardized each regional series across time, and performed no ROI re-binning or dimensional aggregation.

Accordingly, each subject was represented by an AAL116 regional time-series matrix. Let $\mathbf{X} \in \mathbb{R}^{T\times 116}$ denote the regional time-series matrix, where T is the number of temporal observations.

To characterize complementary forms of interregional interaction, three functional connectivity views were constructed for every subject: Pearson correlation (PC), sparse representation (SR), and Granger causality mapping (GCM).

Pearson correlation view.

The PC view characterizes undirected linear synchronization between regional time series. The connectivity value between regions i and j is calculated as

$$\mathbf{A}_{ij}^{\mathrm{PC}} = \left| \frac{\sum_t (x_{ti} - \bar{x}_i)(x_{tj} - \bar{x}_j)}{\sqrt{\sum_t (x_{ti} - \bar{x}_i)^2}\sqrt{\sum_t (x_{tj} - \bar{x}_j)^2}} \right|. \tag{1}$$

Sparse representation view.

The SR view characterizes reconstruction-based dependencies. For each regional signal $\mathbf{x}_i$, the remaining regional signals are used to obtain a sparse reconstruction Tibshirani

(1996):

$$\hat{\boldsymbol{\beta}}_i = \arg\min_{\boldsymbol{\beta}_i} \|\mathbf{x}_i - \mathbf{X}_{-i}\boldsymbol{\beta}_i\|_2^2 + \lambda_{\mathrm{SR}} \|\boldsymbol{\beta}_i\|_1 . \tag{2}$$

The resulting coefficients are placed into a connectivity matrix, symmetrized, and normalized to obtain $\mathbf{A}^{\mathrm{SR}}$. In our implementation, λ_{SR} was set to 0.01.

Granger causality mapping view.

The GCM view characterizes directed temporal predictability Granger (1969); Seth et al. (2015). For a pair of regions, the method measures whether the lagged activity of one region improves prediction of the activity of the other region. The relative reduction in residual variance is used as the directed connectivity score, producing $\mathbf{A}^{\mathrm{GCM}}$. In our implementation, the GCM lag order was set to 1.

Each connectivity matrix was converted into a weighted sparse graph using the same graph-construction procedure across all acquisition sites. The strongest connections satisfying the predefined sparsity criterion were retained for every node. The row-wise connectivity profile of each ROI was used as its node feature, yielding three 116-node functional graphs for every subject.

4 Methodology

4.1 Problem Formulation

Let $\mathcal{D}_{s_1} = \{(\mathbf{x}_i^{s_1}, y_i^{s_1})\}_{i=1}^{n_{s_1}}$ and $\mathcal{D}_{s_2} = \{(\mathbf{x}_i^{s_2}, y_i^{s_2})\}_{i=1}^{n_{s_2}}$ denote the two labeled source domains corresponding to Site20 and Site21. For a selected target site, let $\mathcal{D}_t = \{\mathbf{x}_i^t\}_{i=1}^{n_t}$ denote the complete target-domain dataset, whose diagnostic labels are unavailable during training.

The objective is to learn a feature extractor $F(\cdot)$ and a classifier $C(\cdot)$ using the two labeled source domains and the unlabeled target domain Zhao et al. (2018); Peng et al. (2019). After target-domain adaptation is completed, the concealed target labels are used only to evaluate ACC and AUC.

4.2 Overall Framework

Figure 1 illustrates the proposed framework. It contains five principal components:

1. three view-specific graph attention encoders;
2. pairwise cross-view interaction modules;
3. a dual-stream adaptive fusion module;
4. a post-fusion hyperbolic representation encoder; and
5. classification and multi-source domain-adaptation objectives.

For each subject, the PC, SR, and GCM functional graphs are first processed by separate graph attention branches Veličković et al. (2018). Pairwise interaction modules then model dependencies between PC–SR, PC–GCM, and SR–GCM representations. The original view representations and the cross-view interaction representations constitute two complementary information streams, which are integrated through sample-adaptive fusion.

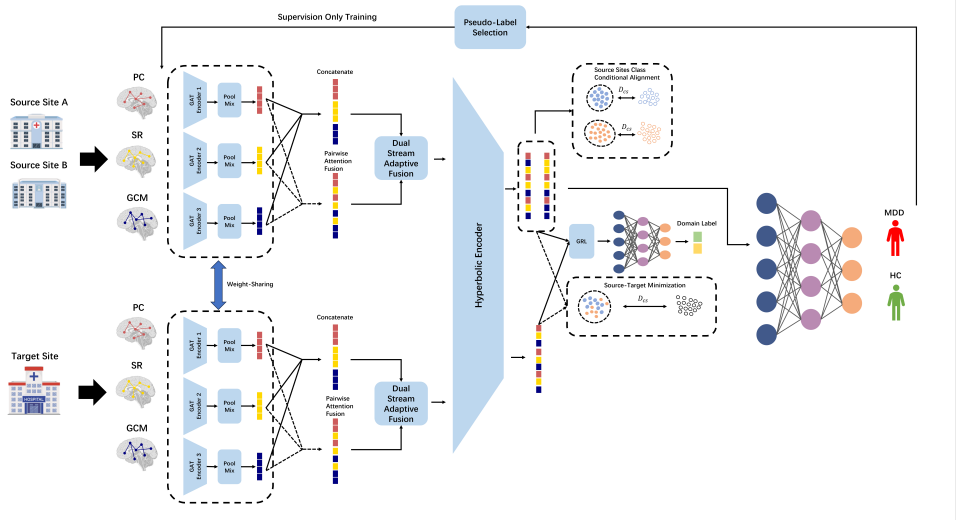


Fig. 1 Overview of the proposed multi-source hyperbolic multi-view graph domain adaptation framework. PC, SR, and GCM functional graphs are encoded by view-specific graph attention branches. Pairwise cross-view interactions and original view representations are integrated by dual-stream adaptive fusion, followed by a post-fusion hyperbolic representation encoder. Source supervision and multi-source domain adaptation are jointly used to optimize the final representation.

The resulting Euclidean fused feature is subsequently processed by a post-fusion hyperbolic encoder. The final representation is used simultaneously for MDD classification, source-source alignment, source-target adaptation, adversarial learning, information maximization, and confidence-aware target pseudo-label supervision.

4.3 View-Specific Graph Attention Encoding

For each connectivity view $v \in \{\mathrm{PC}, \mathrm{SR}, \mathrm{GCM}\}$, a graph attention encoder maps the corresponding node features and weighted graph structure to a graph-level representation:

$$\mathbf{h}_v = f_v\left(\mathbf{X}_v, \mathbf{A}_v; \boldsymbol{\theta}_v\right), \qquad \mathbf{h}_v \in \mathbb{R}^d. \tag{3}$$

Each branch contains graph attention layers, residual feature projection, graph-level mean/max aggregation, dropout, and feature normalization Zaheer et al. (2017). Separate branch parameters are used to preserve the topological characteristics of the three functional connectivity constructions.

4.4 Pairwise Cross-View Interaction

The three view-specific embeddings provide complementary but heterogeneous descriptions of brain functional organization. To model their dependencies explicitly, pairwise interaction features are constructed as

$$\mathbf{q}_{uv} = \mathcal{I}_{uv}\left(\mathbf{h}_u, \mathbf{h}_v\right), \tag{4}$$

where $(u, v) \in \{(\mathrm{PC}, \mathrm{SR}), (\mathrm{PC}, \mathrm{GCM}), (\mathrm{SR}, \mathrm{GCM})\}$ and $\mathcal{I}_{uv}$ denotes the corresponding learnable cross-view attention operator.

The complete interaction stream is

$$\mathbf{q} = [\mathbf{q}_{\mathrm{PC,SR}}; \mathbf{q}_{\mathrm{PC,GCM}}; \mathbf{q}_{\mathrm{SR,GCM}}]. \tag{5}$$

In parallel, the raw-view stream preserves the original view-specific embeddings:

$$\mathbf{r} = [\mathbf{h}_{\mathrm{PC}}; \mathbf{h}_{\mathrm{SR}}; \mathbf{h}_{\mathrm{GCM}}]. \tag{6}$$

4.5 Dual-Stream Adaptive Fusion

The raw-view stream and interaction stream are first projected to a common feature space:

$$\tilde{\mathbf{r}} = \phi_r(\mathbf{r}), \qquad \tilde{\mathbf{q}} = \phi_q(\mathbf{q}). \tag{7}$$

Sample-specific fusion weights are then generated by

$$[\alpha_r, \alpha_q] = \mathrm{softmax}\left(g\left([\tilde{\mathbf{r}}; \tilde{\mathbf{q}}]\right)\right), \tag{8}$$

where $\alpha_r + \alpha_q = 1$. The fused Euclidean representation is

$$\mathbf{z} = \psi\left(\alpha_r \tilde{\mathbf{r}} + \alpha_q \tilde{\mathbf{q}}\right). \tag{9}$$

This design allows the network to adaptively balance direct view-specific information and higher-order cross-view interactions for each subject.

4.6 Post-Fusion Hyperbolic Representation Encoding

Functional brain networks exhibit modular and hierarchical organization, which motivates the use of non-Euclidean representation constraints. After multi-view fusion, we introduce a post-fusion hyperbolic representation encoder based on the Poincaré ball model Nickel and Kiela (2017); Ganea et al. (2018); Chami et al. (2019). Different from applying independent geometric transformations to each view, the encoder operates on the fused multi-view representation and imposes a curvature-controlled constraint on the final subject-level feature.

Let $\mathbf{z} \in \mathbb{R}^d$ denote the fused Euclidean feature, and initialize $\mathbf{h}^{(0)} = \mathbf{z}$. For the ℓ-th hyperbolic encoding layer, a nonlinear tangent-space update is first computed as

$$\mathbf{u}^{(\ell)} = s \cdot \tanh\left(\phi_\ell(\mathbf{h}^{(\ell-1)})\right), \tag{10}$$

where $\phi_\ell(\cdot)$ is a learnable nonlinear projection with layer normalization, ELU activation, dropout, and linear transformation, and s controls the tangent-space scale.

The tangent vector is mapped to the Poincaré ball with curvature $-c$:

$$\tilde{\mathbf{h}}_{\mathbb{D}}^{(\ell)} = \mathrm{Exp}_{\mathbf{0}}^{c}(\mathbf{u}^{(\ell)}). \tag{11}$$

To keep the representation inside the valid Poincaré ball, a curvature-dependent projection is applied:

$$\mathbf{h}_{\mathbb{D}}^{(\ell)} = \mathrm{Proj}^{c}\left(\tilde{\mathbf{h}}_{\mathbb{D}}^{(\ell)}\right), \qquad c\|\mathbf{h}_{\mathbb{D}}^{(\ell)}\|^2 < 1. \tag{12}$$

The projected hyperbolic point is then mapped back to the tangent space for compatibility with the classifier and alignment objectives:

$$\mathbf{h}^{(\ell)} = \mathrm{Log}_{\mathbf{0}}^{c}\left(\mathbf{h}_{\mathbb{D}}^{(\ell)}\right). \tag{13}$$

Finally, a learnable residual gate combines the hyperbolic-encoded representation with the original fused feature:

$$\mathbf{h} = \mathrm{LN}\left(\gamma\mathbf{h}^{(L)} + (1-\gamma)\mathbf{z}\right), \tag{14}$$

where γ is obtained through a sigmoid gate. The resulting feature $\mathbf{h}$ is used for classification, Cauchy–Schwarz alignment, adversarial adaptation, information maximization, and pseudo-label learning.

The intermediate representation is constrained by the curvature-dependent Poincaré-ball projection and subsequently read out in the tangent space for compatibility with the downstream objectives. Together with nonlinear tangent-space transformations and gated residual fusion, the module provides a bounded, curvature-aware refinement of the fused multi-view feature.

4.7 Two-Stage Multi-Source Domain Adaptation

Stage I: supervised source pretraining.

The graph encoders, fusion module, hyperbolic encoder, and classifier are first optimized using labeled samples from Site20 and Site21. The source classification objective is

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{n_{s_1}+n_{s_2}} \sum_{s\in\{s_1,s_2\}} \sum_{i=1}^{n_s} \sum_{c=1}^{C} y_{ic}^{s} \log p_{ic}^{s}. \tag{15}$$

Stage II: unlabeled target-domain adaptation.

During the second stage, the complete target domain participates in adaptation without ground-truth labels. Source classification supervision is retained while source-source and source-target distribution discrepancies are reduced.

4.8 Class-Wise Cauchy–Schwarz Alignment

Let $\mathbf{H}_{20}^{(c)}$ and $\mathbf{H}_{21}^{(c)}$ denote the final features of source samples from class c in Site20 and Site21, respectively. Their class-conditional discrepancy is measured by

$$\mathcal{L}_{ss} = \frac{1}{C} \sum_{c=1}^{C} D_{\mathrm{CS}} \left(\mathbf{H}_{20}^{(c)}, \mathbf{H}_{21}^{(c)} \right). \tag{16}$$

The Cauchy–Schwarz divergence between two feature sets $\mathbf{X}$ and $\mathbf{Y}$ is defined as

$$D_{\mathrm{CS}}(\mathbf{X}, \mathbf{Y}) = -\log \frac{\mathbb{E}_{\mathbf{x},\mathbf{y}}[k(\mathbf{x}, \mathbf{y})]}{\sqrt{\mathbb{E}_{\mathbf{x},\mathbf{x}'}[k(\mathbf{x}, \mathbf{x}')]\, \mathbb{E}_{\mathbf{y},\mathbf{y}'}[k(\mathbf{y}, \mathbf{y}')]}}, \tag{17}$$

where $k(\cdot, \cdot)$ is an RBF kernel Jenssen et al. (2006); Gretton et al. (2012).

For confident target samples, pseudo-labels are used to construct class-specific target feature sets $\mathbf{H}_t^{(c)}$. Class-wise source-target alignment is then defined as

$$\mathcal{L}_{st} = \frac{1}{2C} \sum_{s \in \{20,21\}} \sum_{c=1}^{C} D_{\mathrm{CS}} \left(\mathbf{H}_s^{(c)}, \mathbf{H}_t^{(c)} \right). \tag{18}$$

This formulation preserves disease-related class structure while reducing both inter-source and source-target discrepancies.

4.9 Domain-Adversarial Learning

A binary domain discriminator $G_d(\cdot)$ is connected to the final feature through a gradient reversal layer. The discriminator is trained to distinguish source and target representations, while the feature extractor is trained adversarially to produce domain-invariant representations Ganin et al. (2016); Long et al. (2018). Site20 and Site21 are handled as two source domains for source-source alignment, while the adversarial discriminator operates on source-versus-target discrimination. The adversarial loss is denoted by

$$\mathcal{L}_{\mathrm{adv}} = -\sum_{i} \sum_{d} d_{id} \log \hat{d}_{id}. \tag{19}$$

4.10 Information Maximization

For target prediction $\mathbf{p}_i^t$, the information-maximization objective Grandvalet and Bengio (2004); Liang et al. (2020) is

$$\mathcal{L}_{\mathrm{im}} = \frac{1}{n_t} \sum_{i=1}^{n_t} H\left(\mathbf{p}_i^t\right) - H\left(\frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{p}_i^t \right), \tag{20}$$

where $H(\cdot)$ denotes entropy. Minimizing the first term encourages confident predictions, while maximizing the marginal entropy prevents all target samples from collapsing into the same class.

4.11 Confidence-Aware Pseudo-Label Learning

A target sample is included in pseudo-label supervision only when its maximum predicted probability exceeds a confidence threshold τ Lee (2013); Sohn et al. (2020); Liang et al. (2020). Let

$$\mathcal{I}_\tau = \left\{ i : \max_c p_{ic}^t \geq \tau \right\}. \tag{21}$$

The pseudo-label loss is

$$\mathcal{L}_{\mathrm{pl}} = -\frac{1}{|\mathcal{I}_\tau|} \sum_{i \in \mathcal{I}_\tau} \sum_{c=1}^{C} \hat{y}_{ic}^t \log p_{ic}^t. \tag{22}$$

The confidence threshold and the contribution of pseudo-label supervision are progressively scheduled to reduce the influence of incorrect target predictions during early adaptation.

4.12 Overall Optimization Objective

The complete adaptation objective is

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\mathrm{cls}} &+ \lambda_{ss}\mathcal{L}_{ss} + \lambda_{st}\mathcal{L}_{st} \\ &+ \lambda_{\mathrm{adv}}\mathcal{L}_{\mathrm{adv}} + \lambda_{\mathrm{im}}\mathcal{L}_{\mathrm{im}} + \lambda_{\mathrm{pl}}\mathcal{L}_{\mathrm{pl}}. \end{aligned} \tag{23}$$

The loss weights are dynamically scheduled during adaptation so that source-discriminative information is preserved while target-domain alignment becomes progressively stronger.

5 Experiments

5.1 Comparison Methods and Evaluation Metrics

The proposed method was compared with BC-SVM, LE-SVM, DANN, UFA-Net, MFCP, AUFA, and H2MSDA. BC-SVM and LE-SVM are conventional functional-connectivity pipelines; DANN, UFA-Net, and AUFA are domain-adaptation baselines Long et al. (2015, 2018); MFCP is a multi-view non-adaptation baseline; and H2MSDA models multiple source domains Zhao et al. (2018); Peng et al. (2019). For single-source methods, Site20 only, Site21 only, and pooled Site20+Site21 settings were evaluated. The pooled setting was used in the main comparison to match the number of labeled samples, whereas H2MSDA and the proposed method retained Site20 and Site21 as distinct sources. ACC and AUC were macro-averaged across the seven target sites Fawcett (2006).

5.2 Implementation Details

The model used three view-specific graph attention branches and was optimized with AdamW Loshchilov and Hutter (2019) in two stages: source-supervised pretraining

Table 2 Cross-site target-domain performance across seven sites. Each entry reports ACC/AUC. Single-source baselines use pooled Site20+Site21 data, whereas H2MSDA and the proposed method preserve Site20 and Site21 as distinct source domains. Bold values indicate the highest result for each metric at each target site.

Method	Site6	Site11	Site12	Site16	Site19	Site22	Site25	Mean
UFA-Net	0.6000/0.6670	0.4920/0.6910	0.7110/0.4790	0.5320/0.4920	0.6670/0.6680	0.6000/0.6050	0.6180/0.6000	0.6028/0.6046
MFCP	0.5000/0.7470	0.5740/0.5660	0.4470/0.7710	0.6130/**0.6910**	0.5980/0.6240	0.6000/0.6080	0.4210/0.5000	0.5361/0.6473
BC-SVM	0.5330/0.4620	0.6230/0.5980	0.6320/0.3850	0.5160/0.4480	0.6210/0.5860	0.5800/0.5280	0.5990/0.6000	0.5862/0.5155
LE-SVM	0.5670/0.5200	0.4590/0.4940	0.7370/0.7080	0.5480/0.5860	0.6440/0.5820	0.6400/0.6310	0.6120/0.5000	0.6009/0.5832
DANN	0.5330/0.6000	0.5080/0.6220	0.7890/0.7500	0.5000/0.4860	0.6090/**0.8510**	0.5800/0.6920	0.6050/0.5000	0.5894/0.6472
H2MSDA	0.7000/0.6756	0.6393/0.6175	0.8421/0.7396	0.5806/0.6504	0.6897/0.7783	0.6800/**0.7550**	0.6053/0.4839	0.6767/0.6715
AUFA	0.5000/0.5911	0.5082/0.4871	0.8158/0.5833	0.5000/0.6181	0.5747/0.5507	0.6000/0.6450	0.5987/0.6133	0.5853/0.5841
Ours	**0.7667/0.7511**	**0.6885/0.7026**	**0.8684/0.7760**	**0.6774**/0.6597	**0.7931**/0.8159	**0.7200**/0.6983	**0.6382/0.6294**	**0.7360/0.7190**

followed by unlabeled target-domain adaptation. The pseudo-label confidence threshold was progressively relaxed, and pseudo labels were updated at fixed intervals. A binary discriminator aligned the pooled source and target representations, while class-wise Cauchy–Schwarz losses preserved inter-source and source-target class structure. The model was implemented in PyTorch 2.6.0 Paszke et al. (2019) with CUDA 11.8 and trained on an NVIDIA GeForce RTX 4060 Laptop GPU.

5.3 Main Cross-Site Results

Table 2 presents the complete target-domain performance across the seven acquisition sites.

The proposed method achieved the highest macro-averaged ACC of 73.60% and the highest macro-averaged AUC of 71.90%. Compared with H2MSDA, the strongest multi-source baseline, the proposed method improved mean ACC by 5.93 percentage points and mean AUC by 4.75 percentage points.

The proposed method obtained the highest ACC on all seven target sites. It also achieved the highest AUC on Site6, Site11, Site12, and Site25. MFCP, DANN, and H2MSDA obtained the highest individual AUC on Site16, Site19, and Site22, respectively. This site-dependent behavior indicates that scanner- and cohort-related distribution shifts differ substantially across target domains. Overall, the results suggest that jointly modeling multi-view connectivity, source-domain structure, and target-domain adaptation improves performance across the evaluated unlabeled target sites.

5.4 Source-Setting Sensitivity of Single-Source Methods

To investigate the sensitivity of conventional single-source methods to source-domain selection, each method was evaluated using Site20 only, Site21 only, and Site20+Site21 pooled as one source domain. AUFA was included in this analysis because aggregate results were available for all three source configurations.

The results indicate that single-source methods are highly sensitive to source-domain selection. No method consistently achieved the best performance under all three configurations. Moreover, directly pooling Site20 and Site21 did not uniformly improve performance. This finding suggests that simply increasing the number of labeled source samples does not adequately address inter-source distribution discrepancy.

Table 3 Macro-averaged ACC and AUC of single-source methods under different source-data configurations. Site20+Site21 denotes pooling the two source datasets and treating them as a single source domain.

	Site20 only		Site21 only		Site20+Site21 pooled	
Method	ACC	AUC	ACC	AUC	ACC	AUC
UFA-Net	0.6309	0.4995	0.5487	0.5741	**0.6028**	0.6046
MFCP	**0.6725**	0.6174	0.5281	0.6231	0.5361	**0.6473**
BC-SVM	0.5784	0.5100	0.5746	0.6043	0.5862	0.5155
LE-SVM	0.5930	0.4669	0.6060	0.6287	0.6009	0.5832
DANN	0.5902	**0.6236**	0.5075	**0.6616**	0.5894	0.6472
AUFA	0.5499	0.5366	**0.6246**	0.6421	0.5853	0.5841

Table 4 Source-domain configuration analysis of the proposed architecture.

Source setting	Mean ACC	Mean AUC
Site20 only	0.7012	0.6627
Site21 only	0.6751	0.6017
Site20+Site21	**0.7360**	**0.7190**

5.5 Single-Source and Dual-Source Analysis

The proposed architecture was further evaluated using Site20 only, Site21 only, and the complete Site20+Site21 dual-source configuration. For the single-source variants, the same multi-view and hyperbolic architecture was retained, while the second source domain and inter-source alignment were removed.

The dual-source configuration improved ACC and AUC by 3.48 and 5.63 percentage points over Site20 only, and by 6.09 and 11.73 points over Site21 only. These results support the complementarity of the two source domains and the benefit of preserving their domain identities.

5.6 Ablation Studies

Table 5 evaluates the contributions of domain adaptation, post-fusion hyperbolic representation encoding, and the three connectivity views.

In the naming of the single-view variants, "DA" denotes the complete domain-adaptation objective and "Hyp" denotes the hyperbolic encoder. For example, PC+DA+Hyp retains only the PC connectivity view while preserving both domain adaptation and hyperbolic representation encoding.

Removing domain adaptation reduced mean ACC/AUC from 0.7360/0.7190 to 0.6561/0.5258, while removing the hyperbolic encoder reduced them to 0.6486/0.5456. The strongest single-view variant was SR+Hyp, achieving 0.7034 ACC and 0.6696 AUC, but it remained below the complete model by 3.26 and 4.94 percentage points,

Table 5 Macro-averaged component and single-view ablation results.

Variant	Mean ACC	Mean AUC
Full model	**0.7360**	**0.7190**
No domain adaptation	0.6561	0.5258
w/o hyperbolic encoder	0.6486	0.5456
PC + DA + Hyp	0.6829	0.6372
PC + Hyp	0.6634	0.5881
PC only	0.6638	0.5810
SR + DA + Hyp	0.6606	0.5862
SR + Hyp	0.7034	0.6696
SR only	0.6480	0.5538
GCM + DA + Hyp	0.6407	0.5165
GCM + Hyp	0.6347	0.4974
GCM only	0.6290	0.5085

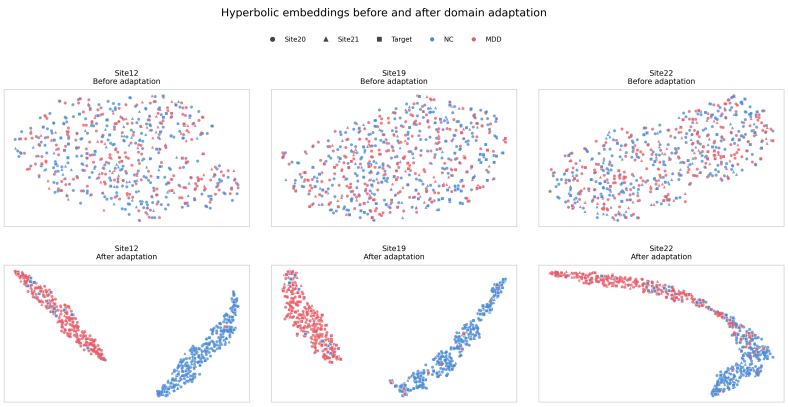


Fig. 2 Joint t-SNE visualization of post-fusion hyperbolic representations before and after domain adaptation for Site12, Site19, and Site22. Colors denote diagnostic classes (blue: HC; red: MDD), while marker shapes denote domains (circle: Site20; triangle: Site21; square: target site). After adaptation, samples from different domains are more consistently organized within class-related manifolds, and the separation between HC and MDD becomes more pronounced.

respectively. Overall, the full three-view configuration achieved the strongest aggregate performance among the evaluated variants.

5.7 Representation Visualization

Figure 2 van der Maaten and Hinton (2008) compares the post-fusion representations before and after target-domain adaptation.

Before adaptation, the embeddings exhibited substantial class overlap and site-dependent variation. After adaptation, they showed clearer class-related organization

across the source and target domains. Because t-SNE is nonlinear, the visualization is treated as qualitative supporting evidence rather than a quantitative measure of alignment.

6 Conclusion

We introduced a unified multi-source, multi-view graph domain adaptation framework for cross-site MDD identification from rs-fMRI. The method preserves view-specific graph structures, explicitly models pairwise cross-view dependencies, adaptively fuses direct and interaction-based representations, and refines the fused feature through lightweight hyperbolic residual encoding. A two-stage transductive adaptation strategy combines class-conditional inter-source and source-target alignment with adversarial learning, information maximization, and confidence-aware pseudo-labeling.

Across seven independently adapted target sites, the framework achieved macro-averaged ACC/AUC of 73.60%/71.90%, exceeding the strongest evaluated baseline by 5.93/4.75 percentage points. Source-configuration and ablation analyses further support the value of preserving source identities, integrating heterogeneous FC views, and applying post-fusion geometric refinement. The current evidence is limited to the evaluated transductive protocol; future work will examine independent-site validation, uncertainty estimation, multimodal integration, and clinically grounded interpretation.

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